

EXPERIMENT NO: 07

Use AFLogical OSE data extract data from an Android device

DATE:

1. AIM:

To configure an Android mobile forensics environment, establish a secure ADB bridge connection, utilize the AFLogical OSE application to perform a logical data extraction, and transfer, verify, and analyze the resulting CSV artifacts.

2. DESCRIPTION:

AFLogical OSE (Open Source Edition, developed by viaForensics) is an application-based tool installed via the Android Debug Bridge (ADB). It queries the device's internal databases to extract critical user artifacts—including contacts, SMS messages, MMS messages, and call logs. The extracted records are compiled and stored locally in standardized CSV (Comma-Separated Values) format, allowing investigators to easily pull, review, and document evidentiary records for investigative reporting.

Logical data extraction is a non-invasive mobile forensics technique used to gather structured data directly from an operating device's active databases and content providers without requiring root access. While physical extraction captures a bit-by-bit clone of the flash memory (enabling the recovery of deleted files), logical extraction interacts exclusively with the operating system's APIs to pull active, allocated data.

3.PROCEDURE:

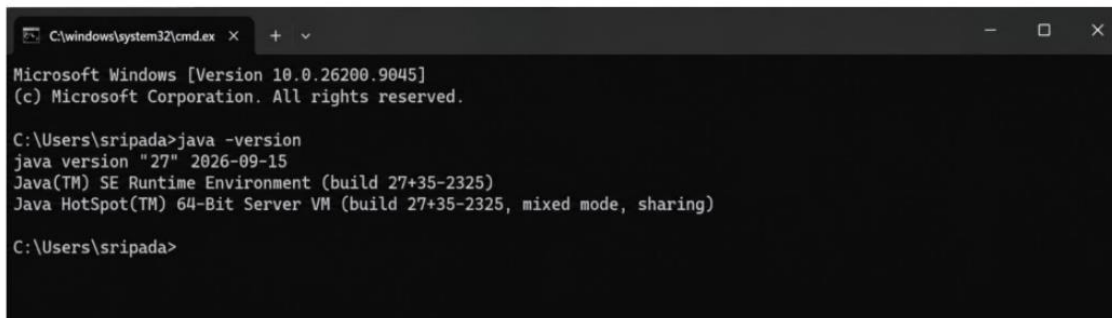
Step 1: Prepare Your Environment 1.

Download AFLLogical OSE:

- o Visit the AFLLogical OSE GitHub page and download the latest release. Alternatively, you can clone the repository using Git.

2. Install Java (if not already installed):

- o AFLLogical OSE requires Java to run. Make sure you have Java installed on your machine. You can download it from the official Java website.



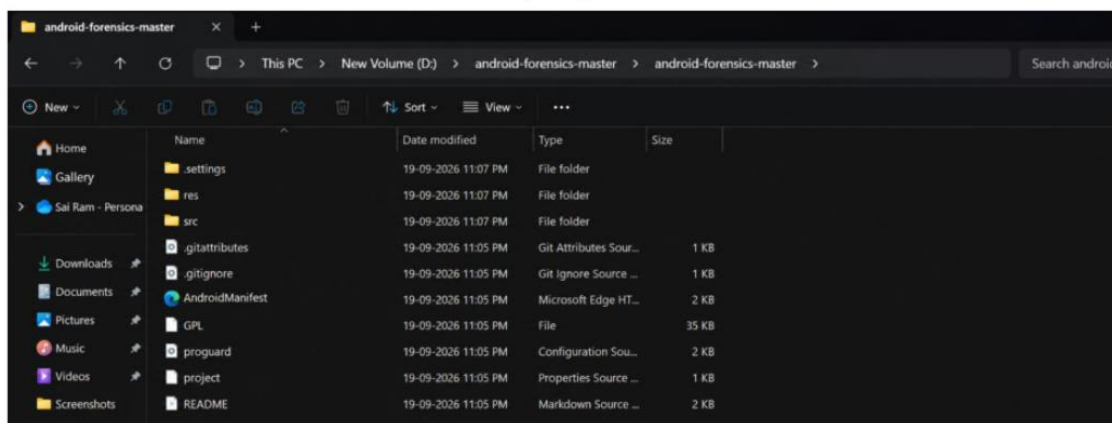
```
C:\windows\system32\cmd.exe x + v
Microsoft Windows [Version 10.0.26200.9045]
(c) Microsoft Corporation. All rights reserved.

C:\Users\sripada>java -version
java version "27" 2026-09-15
Java(TM) SE Runtime Environment (build 27+35-2325)
Java HotSpot(TM) 64-Bit Server VM (build 27+35-2325, mixed mode, sharing)

C:\Users\sripada>
```

3. Install Android Debug Bridge (ADB):

- o ADB is a command-line tool that allows communication with Android devices. Download it from the Android Developer's website and install it on your system.
- o Add ADB to your system's PATH environment variable to easily run ADB commands from the terminal or command prompt



4.Enable USB Debugging on the Android Device:

- o On the Android device, go to Settings > About Phone and tap Build Number seven times to enable Developer Options.
- o Go to Settings > Developer Options and enable USB Debugging.

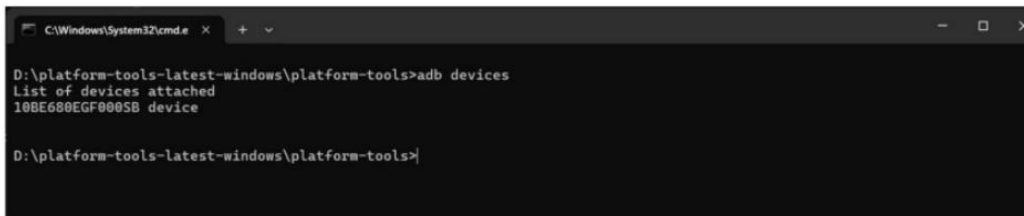
Step 2: Connect the Android Device to Your Computer

1.Connect via USB:

- o Use a USB cable to connect the Android device to your computer.

2.Verify ADB Connection:

- o Open Command Prompt (Windows) or Terminal (macOS/Linux) and run:
Bash
adb devices
- o This command lists connected devices. If your device is listed, you are ready to proceed. If not, ensure that USB Debugging is enabled and that the correct drivers are installed.



```
C:\Windows\System32\cmd.exe x + -
D:\platform-tools-latest-windows\platform-tools>adb devices
List of devices attached
10BE680EGF0005B device

D:\platform-tools-latest-windows\platform-tools>
```

Step 3: Extract Data Using AFLogical OSE

1.Run AFLogical OSE:

- oNavigate to the directory where you have downloaded or extracted AFLogical OSE.

2.Push the APK to the Android Device:

- oRun the following command to install the AFLogical OSE APK on the device:

bash

adb install aflogical.apk

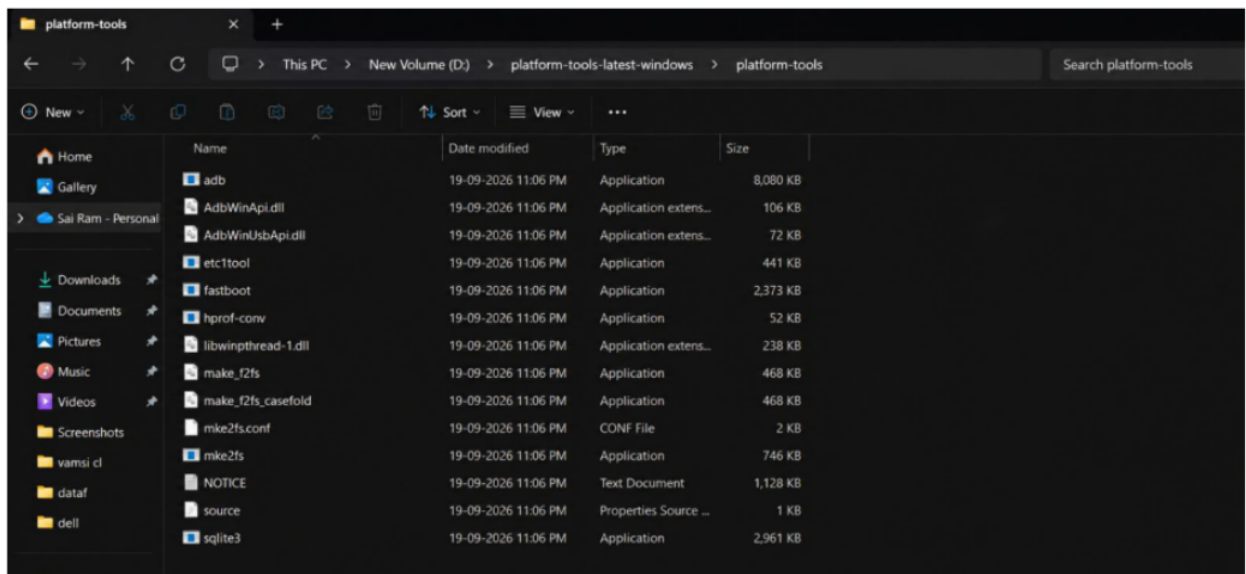
- oThis installs the AFLogical app on the device, which will be used to collect data.

3.Launch the AFLogical OSE App on the Device:

- oOn the Android device, open the AFLogical app.
- oYou'll see options to extract various types of data (e.g., contacts, SMS, MMS, call logs). Select the data types you want to extract.

4.Start Data Extraction:

- oOnce you've selected the data types, the app will start the extraction process.
- oData is stored in .csv files on the device's storage, typically in a directory named aflogical.



Step 4: Transfer Extracted Data to Your Computer

1. Use ADB to Pull Data:

- o After extraction, you need to transfer the data files from the Android device to your computer. Run:

bash

adb pull /sdcard/aflogical /path/to/destination

- o Replace /path/to/destination with the directory path where you want to save the extracted data on your computer.

2. Verify the Data:

- o Navigate to the destination directory and check the .csv files to ensure all required data has been extracted.

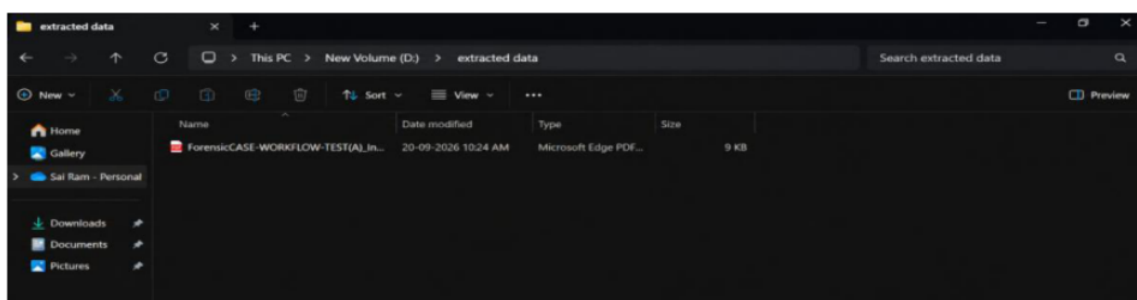
Step 5: Analyze the Extracted Data

1. Open the CSV Files:

- o Use Excel, Google Sheets, or any text editor to open and analyze the .csv files. These files contain the extracted data like contacts, SMS, MMS, and call logs.

2. Review and Document:

- o Carefully review the data for any relevant information. Document your findings and consider exporting the data into a report format for further analysis.



Step 6: Clean Up

1. Uninstall AFLogical OSE:

oOnce data extraction is complete, you can uninstall the AFLogical OSE app from the Android device by running:

```
bash
adb uninstall com.viaforensics.android.aflogical
```

2. Disconnect the Device:

oSafely disconnect the Android device from your computer.
By following these steps, you can use AFLogical OSE to effectively extract and analyze data from an Android device. This process is valuable for forensic investigations, especially when dealing with logical data that doesn't require root access to the device.

4.RUBRICS

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

5. RESULT

AFLogical OSE was successfully used to perform a logical extraction of data from the Android device. The extracted contacts, SMS, MMS, and call logs were transferred to the computer in CSV format and verified for further forensic analysis.